SN65HVD230 vs. TJA1050
Comparison of Responses to CAN Faults

Industrial Interface

May 31st, 2013

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Faults Tested

- The following faults will be tested on a network of two SN65HVD230 devices and on a network of two TJA1050 devices. The results will be compared.
  - Interruption of CAN_H (wire cut)
  - Interruption of CAN_L (wire cut)
  - Short circuit of CAN_H with Vcc
  - Short circuit of CAN_L with GND
  - Short circuit of CAN_H with GND
  - Short circuit of CAN_L with Vcc
- The screen shots were all taken with TXD from the transmitting board and CANH, CANL and RXD on the receiving board.

Note: The TJA1050 is a 5V CAN XCVR and the SN65HVD230 is a 3.3 V XCVR

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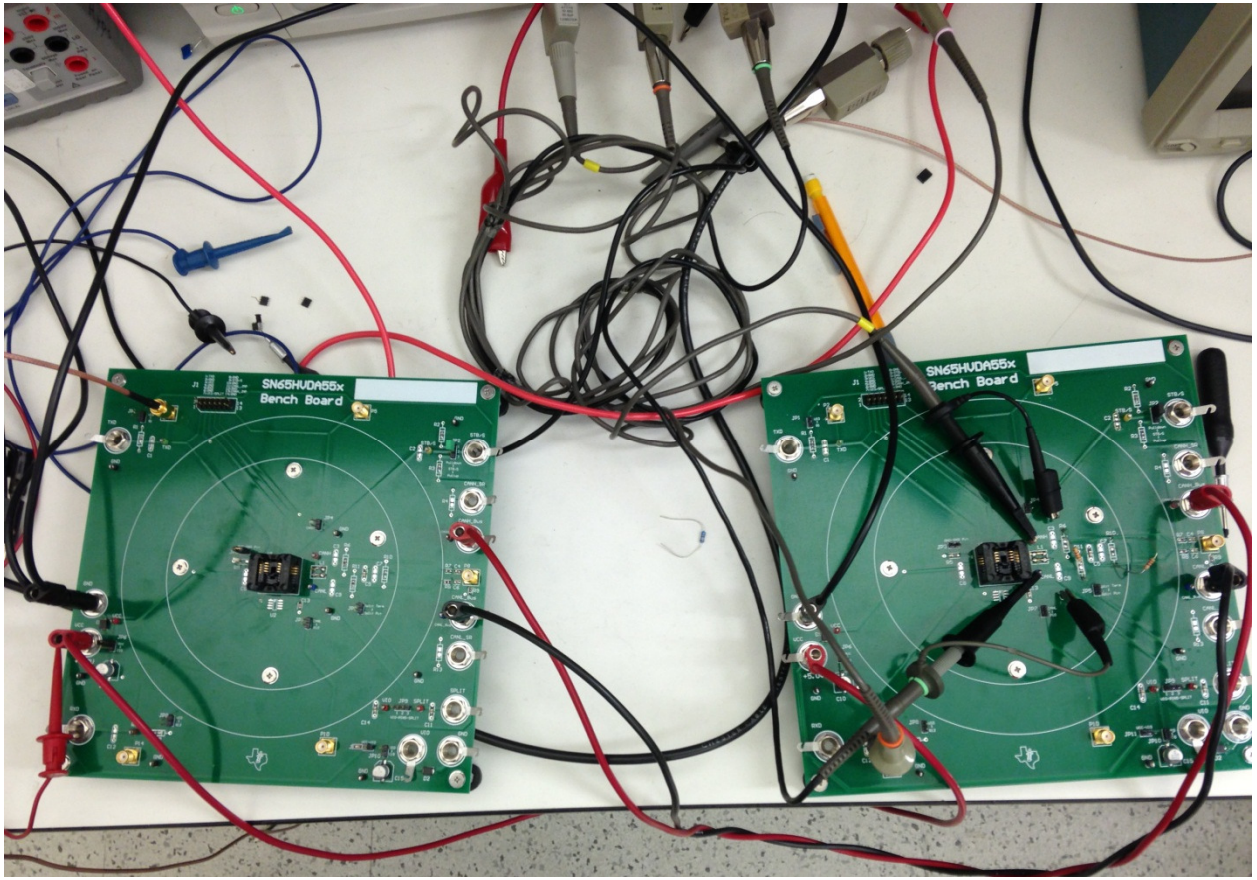
Test Set-Up

The test set-up include two CAN test boards with 2’ of twisted banana plug cabling connecting the CAN bus.



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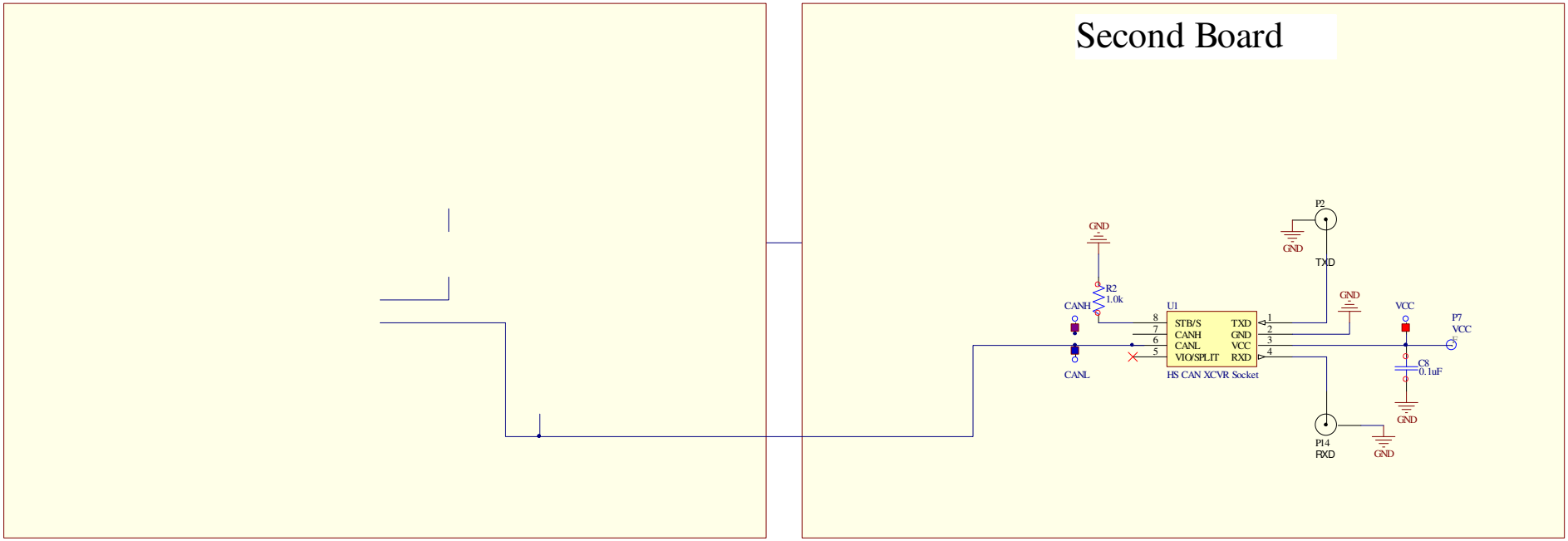
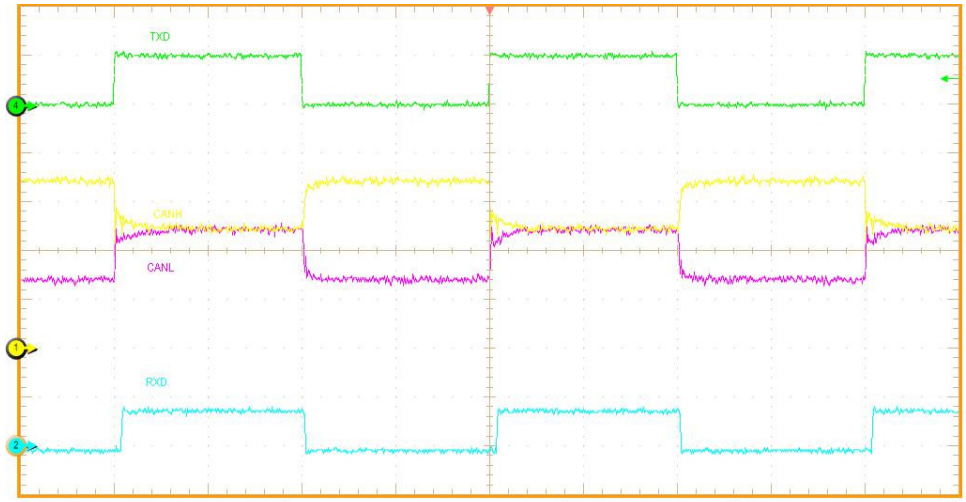
Schematic

Below is a schematic of the components that were populated on the two test boards.



Normal Operation without faults

TJA1050 – Normal



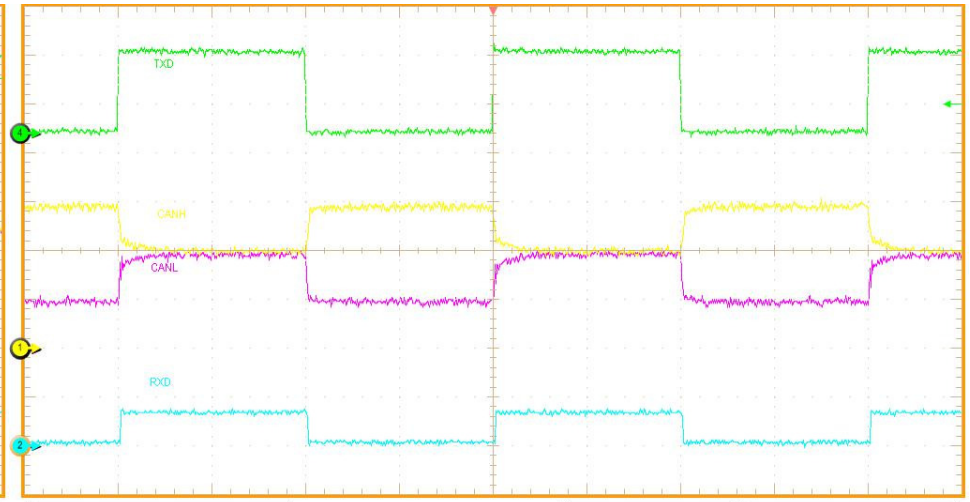
C1	1.0V	Offset:-600mV	1MΩ	B_W:500M			
C2	5.0V/div		1MΩ	B_W:500M			
C3	1.0V/div		1MΩ	B_W:500M			
C4	5.0V/div		1MΩ	B_W:500M			

C4 2.8V

5.0µs	20.0MS/s	50.0ns/pt
Run	Sample	
18 226 acqs		RL:1.0k
May 31, 2013		14:31:40

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	2.473V	2.481535	250.4m	3.554	14.46m	18.24k	
C3	RMS	1.994V	1.9893951	37.35m	3.335	2.754m	37.52k	
M1	RMS*	1.629V	1.6327306	115.1m	5.09	8.515m	54.02k	

SN65HVD230 – Normal



C1	1.0V	Offset:-600mV	1MΩ	B_W:500M			
C2	5.0V/div		1MΩ	B_W:500M			
C3	1.0V/div		1MΩ	B_W:500M			
C4	2.0V/div		1MΩ	B_W:500M			

C4 1.2V

5.0µs	20.0MS/s	50.0ns/pt
Run	Sample	
1 246 acqs		RL:1.0k
May 31, 2013		16:11:07

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	2.425V	2.4248005	236.1m	4.579	6.237m	90.53k	
C3	RMS	1.516V	1.5146526	37.46m	2.319	2.514m	66.35k	
M1	RMS*	1.152V	1.1601632	115.1m	5.09	7.501m	126.3k	

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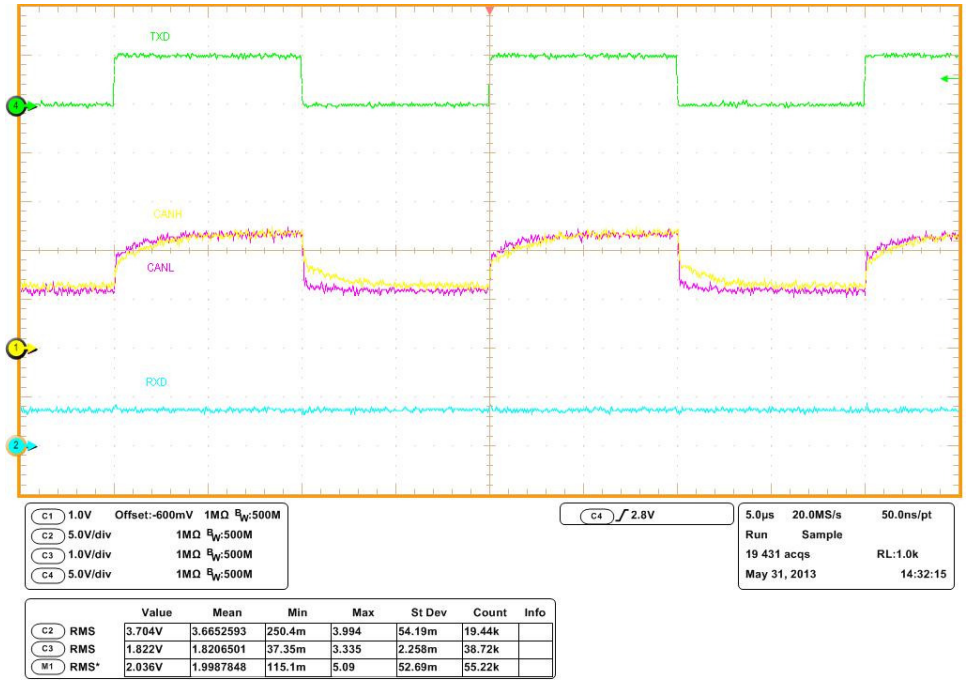
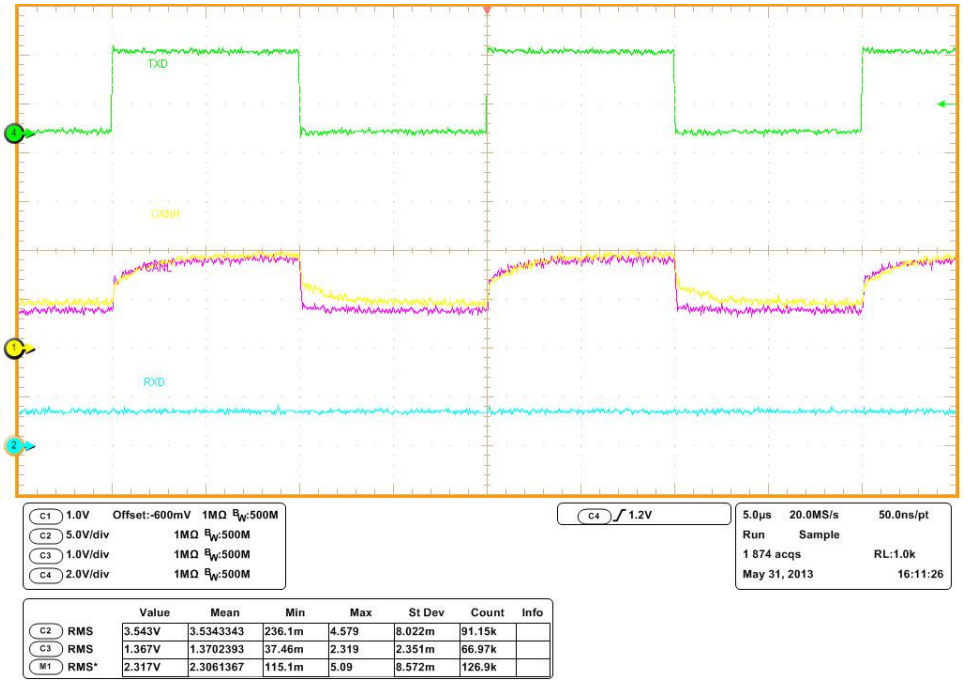
CANH Open on Transmit Board

TJA1050 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	3.704V	3.6652593	250.4m	3.994	54.19m	19.44k	
C3 RMS	1.822V	1.8206501	37.35m	3.335	2.258m	38.72k	
M1 RMS*	2.036V	1.9987848	115.1m	5.09	52.69m	55.22k	

SN65HVD230 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	3.543V	3.5343343	236.1m	4.579	8.022m	91.15k	
C3 RMS	1.367V	1.3702393	37.46m	2.319	2.351m	66.97k	
M1 RMS*	2.317V	2.3061367	115.1m	5.09	8.572m	126.9k	

Conclusion: Neither XCVR is able to communicate with the CANH Bus line open

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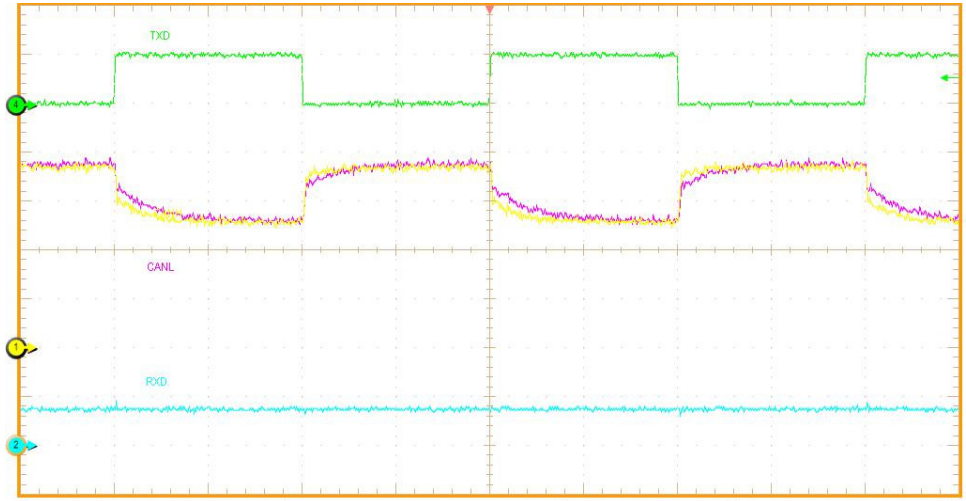
CANL Open on Transmit Board

TJA1050 – Fault



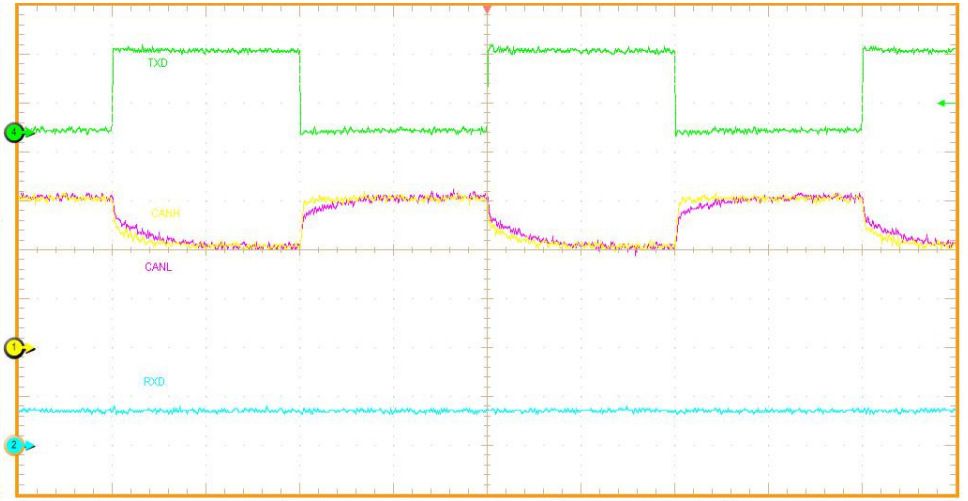
C1 1.0V Offset:-600mV 1MΩ BW:500M
C2 5.0V/div 1MΩ BW:500M
C3 1.0V/div 1MΩ BW:500M
C4 5.0V/div 1MΩ BW:500M

C4 2.8V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
21 011 acqs RL:1.0k
May 31, 2013 14:33:06

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	3.703V	3.7178537	250.4m	3.994	22.27m	21.02k	
C3	RMS	3.276V	3.2778488	37.35m	3.335	2.831m	40.3k	
M1	RMS*	690.0mV	697.21623m	115.1m	5.09	15.21m	56.8k	

SN65HVD230 – Fault



C1 1.0V Offset:-600mV 1MΩ BW:500M
C2 5.0V/div 1MΩ BW:500M
C3 1.0V/div 1MΩ BW:500M
C4 2.0V/div 1MΩ BW:500M

C4 1.2V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
2 341 acqs RL:1.0k
May 31, 2013 16:11:41

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	3.53V	3.5346116	236.1m	4.579	7.905m	91.62k	
C3	RMS	2.666V	2.6661435	37.46m	2.674	3.953m	67.44k	
M1	RMS*	1.005V	1.0107154	115.1m	5.09	7.316m	127.4k	

Conclusion: Neither XCVR is able to communicate with the CANL Bus line open

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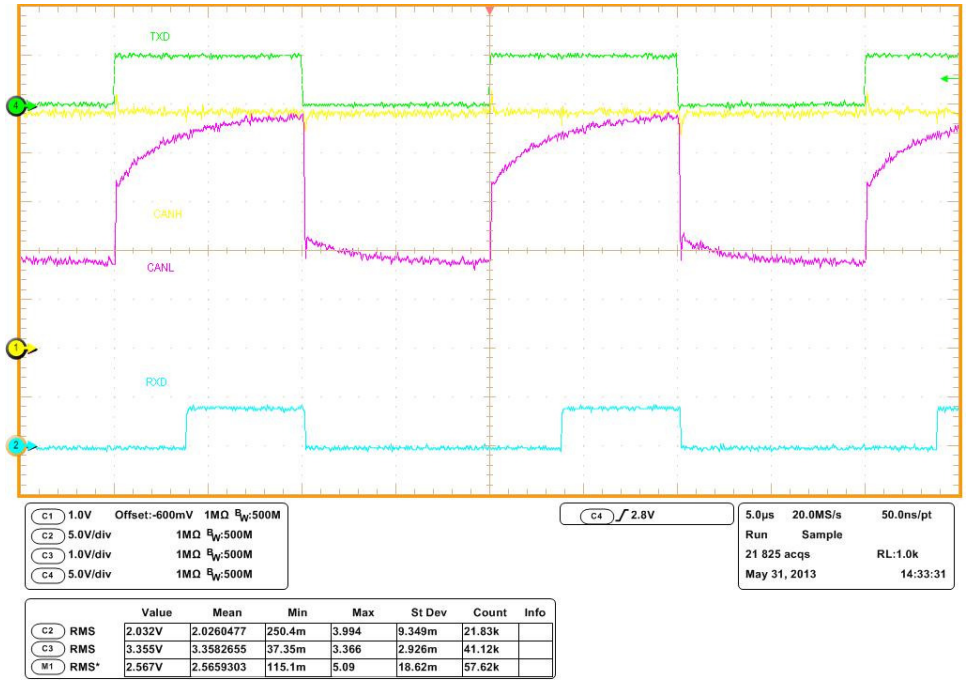
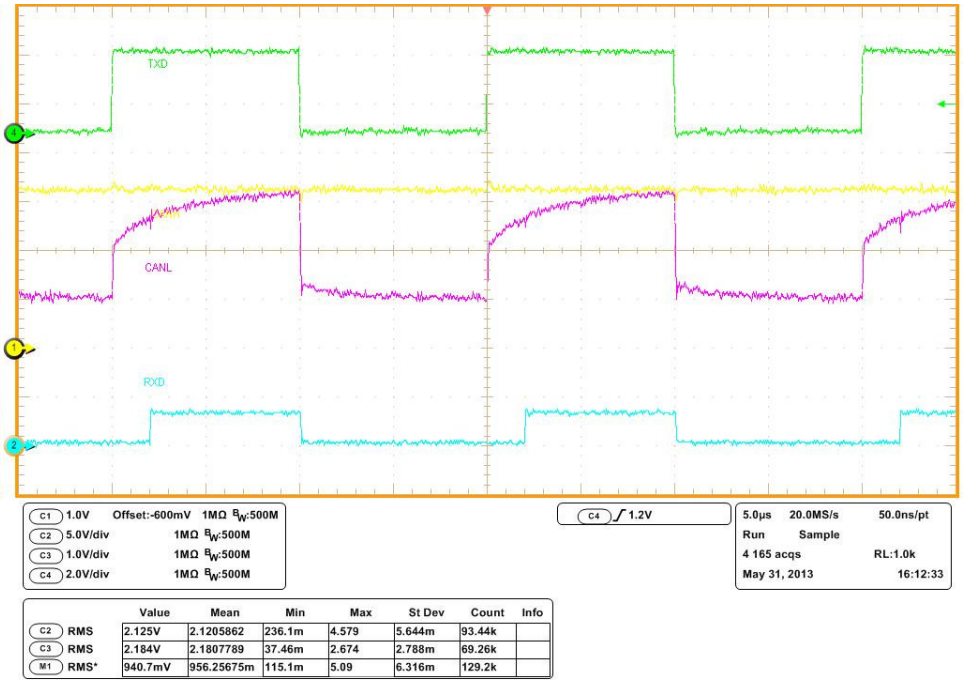
CANH Short to Vcc on Transmit Board

TJA1050 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	2.032V	2.0260477	250.4m	3.994	9.349m	21.83k	
C3 RMS	3.355V	3.3582655	37.35m	3.366	2.926m	41.12k	
M1 RMS*	2.567V	2.5659303	115.1m	5.09	18.62m	57.62k	

SN65HVD230 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	2.125V	2.1205862	236.1m	4.579	5.644m	93.44k	
C3 RMS	2.184V	2.1807789	37.46m	2.674	2.788m	69.26k	
M1 RMS*	940.7mV	956.25675m	115.1m	5.09	6.316m	129.2k	

Conclusion: Both transceivers are marginally able to receive bits with skewed bit timing during a CANH short to Vcc fault.

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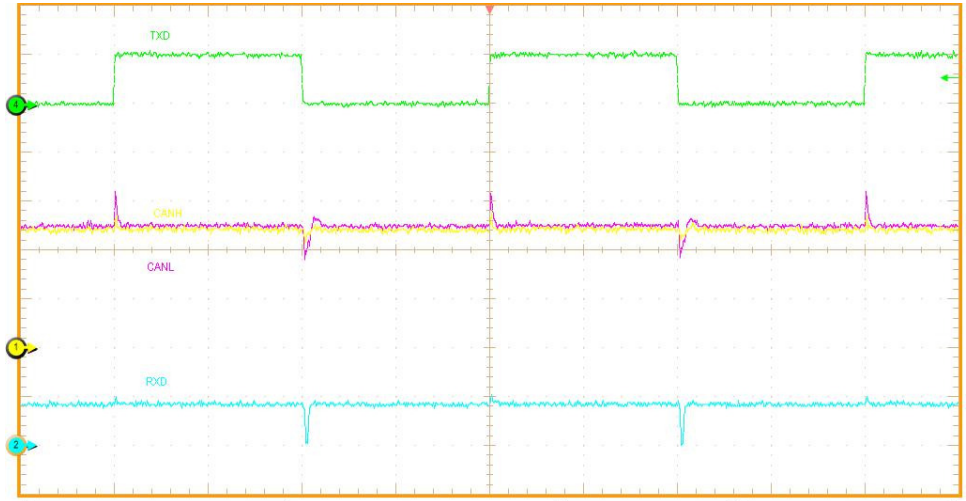
CANL Short to Vcc on Transmit Board

TJA1050 – Fault



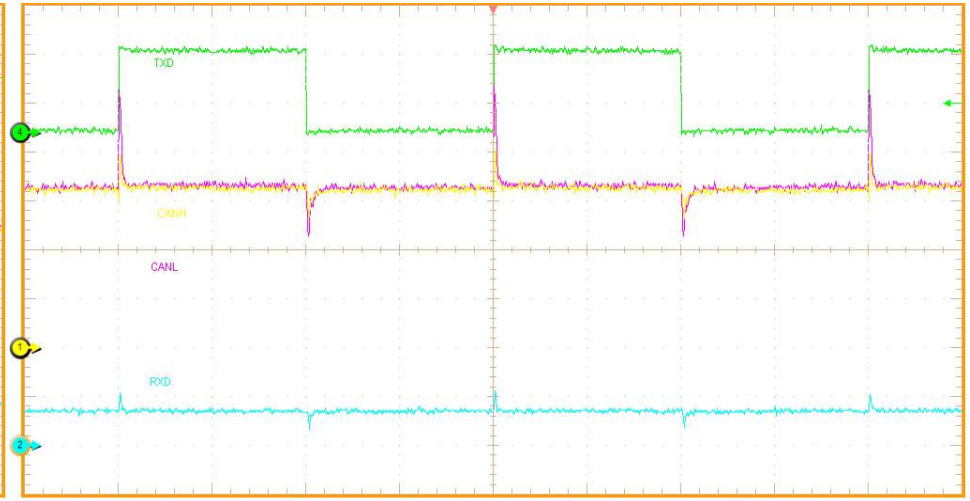
C1	2.0V	Offset:-600mV	1MΩ	BW:500M			
C2	5.0V/div		1MΩ	BW:500M			
C3	2.0V/div		1MΩ	BW:500M			
C4	5.0V/div		1MΩ	BW:500M			

C4 2.8V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
1 265 acqs RL:1.0k
May 31, 2013 14:34:33

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	4.304V	4.2614282	250.4m	4.4	26.67m	23.83k	
C3	RMS	5.023V	5.0220366	2.039	5.032	3.185m	1.26k	
M1	RMS*	786.2mV	827.27306m	115.1m	5.09	25.81m	59.61k	

SN65HVD230 – Fault



C1	1.0V	Offset:-600mV	1MΩ	BW:500M			
C2	5.0V/div		1MΩ	BW:500M			
C3	1.0V/div		1MΩ	BW:500M			
C4	2.0V/div		1MΩ	BW:500M			

C4 1.2V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
4 513 acqs RL:1.0k
May 31, 2013 16:12:45

		Value	Mean	Min	Max	St Dev	Count	Info
C2	RMS	3.532V	3.5306104	236.1m	4.579	16.01m	93.8k	
C3	RMS	3.33V	3.3308326	37.46m	3.338	17.99m	69.61k	
M1	RMS*	259.9mV	257.78753m	115.1m	5.09	12.47m	129.6k	

Conclusion: Neither XCVR is able to communicate with the CANL Short to Vcc

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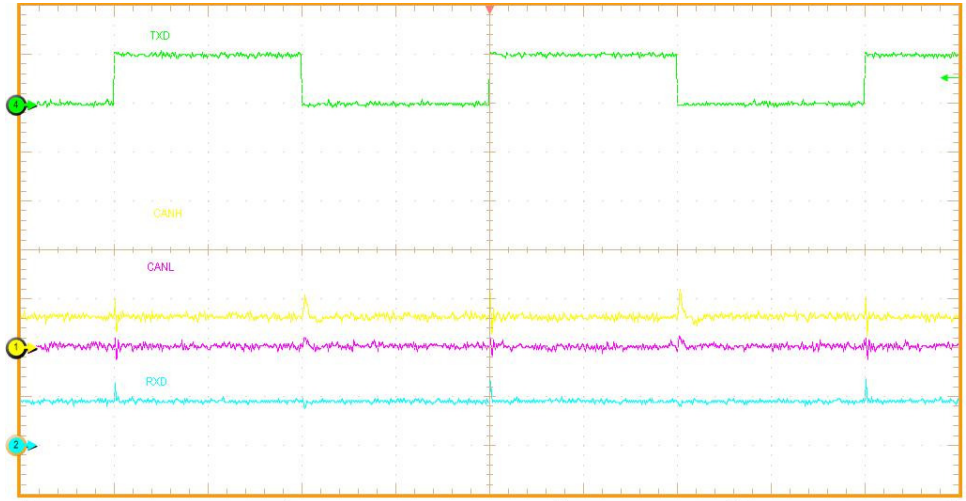
CANH Short to GND on Transmit Board

TJA1050 – Fault



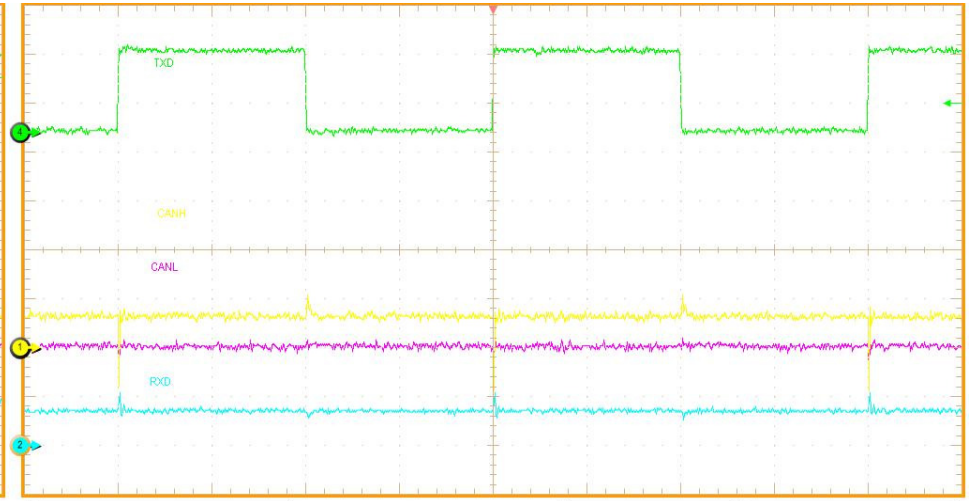
C1	1.0V	Offset:-600mV	1MΩ	B_W:500M			
C2	5.0V/div		1MΩ	B_W:500M			
C3	1.0V/div		1MΩ	B_W:500M			
C4	5.0V/div		1MΩ	B_W:500M			

C4 2.8V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
931 acqs RL:1.0k
May 31, 2013 14:35:12

	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	4.487V	4.4417572	250.4m	4.579	43.52m	25.12k	
C3 RMS	83.44mV	74.267402m	61.79m	88.31m	6.728m	934.0	
M1 RMS*	4.439V	4.3932646	115.1m	5.09	43.4m	60.9k	

SN65HVD230 – Fault



C1	1.0V	Offset:-600mV	1MΩ	B_W:500M			
C2	5.0V/div		1MΩ	B_W:500M			
C3	1.0V/div		1MΩ	B_W:500M			
C4	2.0V/div		1MΩ	B_W:500M			

C4 1.2V

5.0µs 20.0MS/s 50.0ns/pt
Run Sample
5 222 acqs RL:1.0k
May 31, 2013 16:13:07

	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	3.542V	3.5319713	236.1m	4.579	6.587m	94.51k	
C3 RMS	69.36mV	65.681375m	37.46m	3.338	2.175m	70.32k	
M1 RMS*	3.489V	3.4830102	115.1m	5.09	6.169m	130.3k	

Conclusion: Neither XCVR is able to communicate with the CANH short to GND

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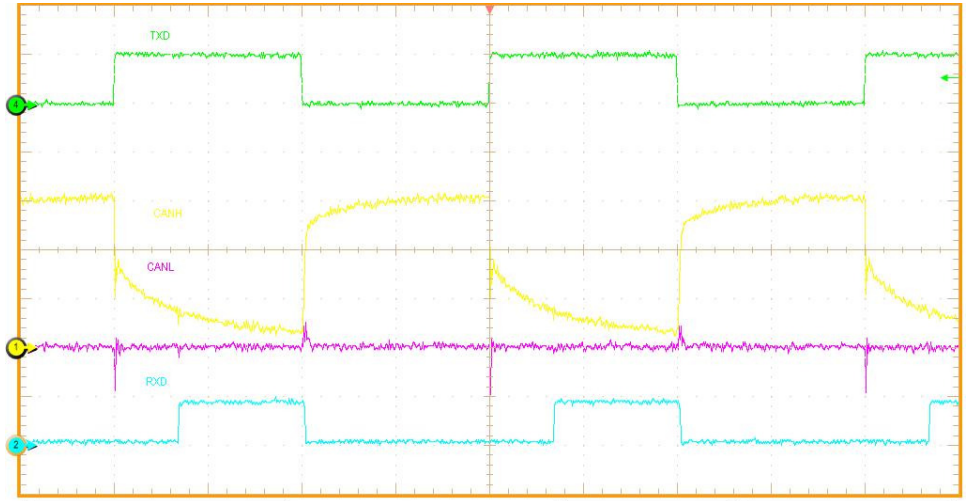
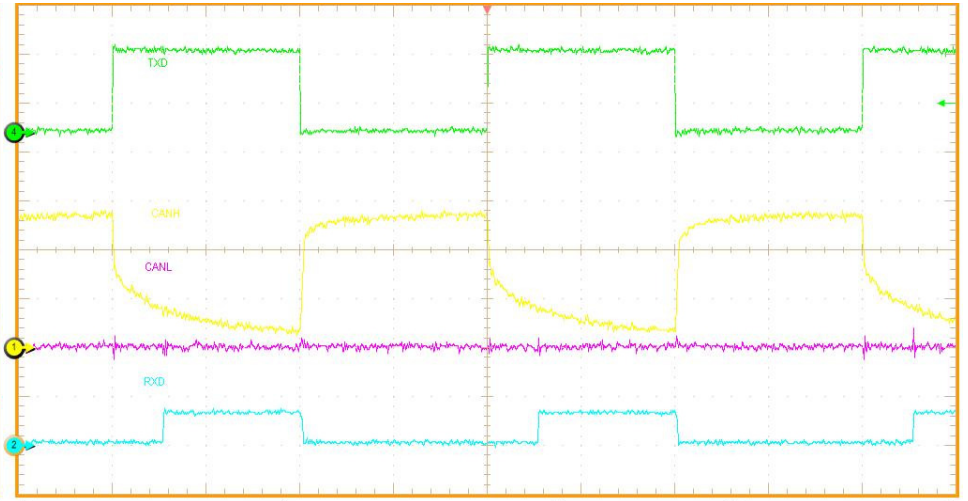
CANL Short to GND on Transmit Board

TJA1050 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	2.411V	2.4319695	250.4m	4.579	11.41m	26.22k	
C3 RMS	69.99mV	65.733154m	53.92m	1.999	7.812m	2.038k	
M1 RMS*	2.39V	2.4095401	115.1m	5.09	11.28m	62.0k	

SN65HVD230 – Fault



	Value	Mean	Min	Max	St Dev	Count	Info
C2 RMS	1.998V	2.0013183	236.1m	4.579	6.245m	95.14k	
C3 RMS	54.12mV	57.96866m	37.46m	3.338	1.813m	70.96k	
M1 RMS*	1.977V	1.9763432	115.1m	5.09	6.724m	130.9k	

Conclusion: Both transceivers are marginally able to receive bits with skewed bit timing during a CANL short to GND fault.

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Conclusion

- In 4 fault cases neither transceiver pair is able to communicate
- In 2 fault cases both pairs of transceivers are able to marginally communicate
  - Both transceiver receive bits that are skewed

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Thank You!